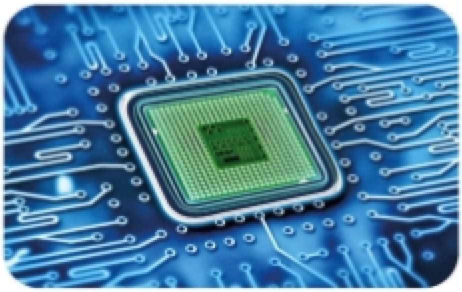
# Silicon

14
**Si**

The second-most abundant element in Earth's crust, silicon forms most rocks. Its compound silicon dioxide is one of the major components of sand. Pure silicon is a bluish-gray solid. It's a semiconductor, which means it doesn't conduct electricity as well as a metal, but it's also not an insulator.

Laboratory sample of pure silicon


Silicon-based microchip

## Microchips

Many modern machines work on microchips. A microchip stores and processes data and is made from a tiny wafer of silicon—a material that allows electrical signals to be transmitted precisely and quickly.


Plastic bag

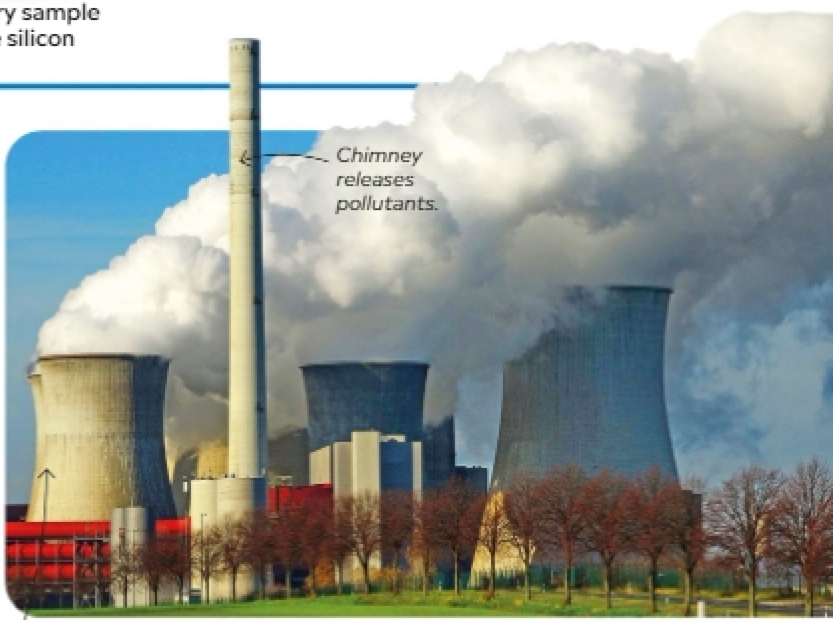
Plastic box

## Powerful plastic

Plastic is an artificial material made up of long chains of carbon atoms. Chemists can combine other elements—such as hydrogen, oxygen, fluorine, or chlorine—with carbon to manufacture different types of plastic for a variety of applications. Water resistant and relatively cheap, plastics have become very widely used. However, recently people have become worried about plastic pollution, and there have been moves to reduce the amount of single-use plastic we use.

## Fossil fuels

Coal, natural gas, and oil are carbon-based fossil fuels—fuels formed in Earth's crust from the remains of ancient organisms. They form the basis of the world's energy supply, but concerns over pollution from burning them have led to the growing use of alternative energy sources.




## Light and sturdy

A lightweight, ultra-strong, human-made material called carbon fiber is used in the aerospace and car industries, where it can replace much heavier materials, such as steel. Race cars made of carbon fiber components are lighter and faster than earlier cars.